Nusrat Binta Nizam

Ph.D. Candidate, Biomedical Engineering | Cornell University & Cornell Tech | New York, NY

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Research Interests

Multimodal Learning • Computer Vision • Vision–Language Models • Medical Image Analysis • Representation Learning • Reinforcement Learning • Optimization

Education

Cornell University & Cornell Tech, New York, NY — Aug 2023 – Dec 2027 (exp.)
Ph.D. in Biomedical Engineering | Advisor: Prof. Mert Rory Sabuncu
Focus: Multimodal learning and personalized screening

Cornell University & Cornell Tech, New York, NY — Aug 2023 – May 2026
M.Sc. in Biomedical Engineering | Advisor: Prof. Mert Rory Sabuncu
Focus: AI-driven opportunistic screening

Bangladesh University of Engineering and Technology, Bangladesh — Feb 2016 – July 2023
B.Sc. & M.Sc. in Biomedical Engineering | Advisor: Dr. Taufiq Hasan
Focus: Biomedical signal processing

Research Experience

Cost-Aware Diagnostic Pathway Optimization — May 2026 – Present
Cornell University & Cornell Tech | Advisor: Prof. Mert Rory Sabuncu

– Developing a reinforcement learning agent that selects the next diagnostic test under cost and accuracy constraints, producing patient-specific screening pathways.

Robust Multimodal Learning with Missing Modalities — Oct 2025 – May 2026
Cornell University & Cornell Tech | Advisor: Prof. Mert Rory Sabuncu

– Built a Shapley-adaptive modality knockout scheme that modulates training when one or more modalities are absent, improving robustness over standard fusion baselines.

Multimodal AI for Cardiac Amyloidosis Screening — Aug 2024 – Feb 2026
Cornell Tech & Weill Cornell Medicine | Advisors: Prof. Mert Rory Sabuncu, Dr. Ashley Beecy

– Built a multimodal deep learning system fusing CT imaging, ECG, and structured clinical features to detect ATTR cardiac amyloidosis in patients with severe aortic stenosis.

Opportunistic Cardiac Screening from Routine Chest CT — Aug 2023 – April 2026
Cornell Tech | Advisor: Prof. Mert Rory Sabuncu

– Developed a deep learning framework that detects cardiac abnormalities from non-cardiac chest CT scans acquired for unrelated indications.

Cross-Modal Cardiac Knowledge Transfer — Jan 2025 – Dec 2025
Cornell Tech | Advisor: Prof. Mert Rory Sabuncu

– Designed a phenotype-guided contrastive pre-training strategy aligning imaging and clinical text for vision–language models.

COVID-19 Severity Prediction from Chest X-rays — Jun 2021 – Jun 2023
mHealth Lab, BUET | Advisor: Dr. Taufiq Hasan

– Proposed an anatomy-aware convolutional network for severity classification.

RadAssist: AI-Powered Tele-Radiology Platform — May 2020 – Jun 2022
mHealth Lab, BUET | Advisor: Dr. Taufiq Hasan

- Built a full-stack web application integrating multiple AI models for automated report generation and disease detection, with a DICOM processing pipeline.

Noise-Robust Heart Disease Classification May 2019 – Feb 2021
mHealth Lab, BUET | Advisor: Dr. Taufiq Hasan

- Developed Hilbert-envelope features for phonocardiogram analysis, reaching 91% accuracy on valvular heart disease classification from noisy recordings.

Technical Skills

Methods	Deep Learning, Computer Vision, Vision–Language Models, Multimodal Learning, Reinforcement Learning, Natural Language Processing, Agentic AI, Optimization
Languages	Python, C++, MATLAB, Bash, LaTeX
Frameworks	PyTorch, TensorFlow, Keras, OpenCV, scikit-learn, Pandas, NumPy
Infrastructure	CUDA, Git, Azure, HPC/SLURM clusters, DICOM pipelines

Professional Experience

Graduate Research Assistant, Cornell University Jan 2024 – Present
New York, NY

Lecturer (on study leave), Dept. of Biomedical Engineering Aug 2021 – July 2023
Bangladesh University of Engineering and Technology, Dhaka, Bangladesh

Undergraduate Researcher, mHealth Lab May 2019 – May 2021
Bangladesh University of Engineering and Technology, Dhaka, Bangladesh

Industrial Trainee, SIEMENS Apr 2019 – May 2019
Dhaka, Bangladesh

Publications

1. **Nizam, N.B.**, Liu, F., Kwak, S., ... & Sabuncu, M. ShapKO: Shapley-Adaptive Modality Knockout for Robust Multimodal Learning. *MICCAI*, 2026. [Paper]
2. **Nizam, N.B.**, Beecy, A., Groenendyk, J., Liu, F., ... & Sabuncu, M. Cardio Amyloid-AI: Advanced Multimodal Screening for Transthyretin Cardiac Amyloidosis in Severe Aortic Stenosis Patients. *European Heart Journal – Digital Health*, 2026. [Paper]
3. **Nizam, N.B.**, Liu, F., Kwak, S., ... & Sabuncu, M. X-Cardia: Phenotype-Guided Cross-Modal Alignment for Opportunistic Cardiac Screening on Routine Chest CT. *MIDL*, 2026. **(Oral, top 8%)** [Paper]
4. Liu, F., Kwak, S., Phung, H., **Nizam, N.B.**, ... & Sabuncu, M. HyperCT: Low-Rank Hypernet for Unified Chest CT Analysis. *MIDL*, 2026. [Paper]
5. Kwak, S., Liu, F., **Nizam, N.B.**, ... & Sabuncu, M. Beyond Machine Interpretation: Learning from Expert Over-Reads Improves ECG Diagnosis. *MIDL*, 2026. [Paper]
6. Liu, F., Kwak, S., **Nizam, N.B.**, ... & Sabuncu, M. RNED: Rotary Number Encoding and Decoding for Quantitative Medical VLM Analysis. *CVPR*, 2026. [Paper]
7. Ali, S.N., Zahin, A., Shuvo, S.B., **Nizam, N.B.**, ... & Hasan, T. BUET Multi-disease Heart Sound Dataset: A Comprehensive Auscultation Dataset for Developing Computer-Aided Diagnostic Systems. *Computer Methods and Programs in Biomedicine Update*, 2026. [Paper]
8. Raikhelkar, J., Bai, Z., Beecy, A., Ilan, R., Liu, F., **Nizam, N.B.**, Kishore, V., ... & Uriel, N. An Artificial Intelligence Model to Detect Abnormal Ejection Fraction from Non-Contrast Chest Computed Tomography: The CT-LVEF Study. *European Heart Journal – Digital Health*, 2026. [Paper]
9. **Nizam, N.B.**, Siddiquee, S.M., Shirin, M., Bhuiyan, M.I.H., & Hasan, T. COVID-19 Severity Prediction from Chest X-ray Images using an Anatomy-Aware Deep Learning Model. *J. Digit. Imaging*, 2023. [Paper]
10. **Nizam, N.B.**, Nuhash, S.I.S.K., & Hasan, T. Hilbert-Envelope Features for Cardiac Disease Classification from Noisy Phonocardiogram. *Biomedical Signal Processing and Control*, 2022. [Paper]
11. Kamal, U., Zunaed, M., **Nizam, N.B.**, & Hasan, T. Anatomy X-Net: A Semi-Supervised Anatomy Aware

Convolutional Neural Network for Thoracic Disease Classification. *IEEE Journal of Biomedical and Health Informatics*, 2022. [Paper]

12. Rahim, R.R., Al Nahian, A.I., Kalam, R.B., Gupta, A.S., & **Nizam, N.B.** Bangla Speech-to-Braille Interaction Device for Visual and Hearing Impaired. *Springer FRAB Conference*, 2022. [Paper]
13. Rahman, M.L., **Nizam, N.B.**, Datta, P., Hasan, M.M., Hasan, T., & Bhuiyan, M.I.H. A Wavelet-CNN Feature Fusion Approach for Detecting COVID-19 from Chest Radiographs. *IEEE ICECE*, 2020. [Paper]
14. **Nizam, N.B.**, Jinan, T., Aurthy, W.B.N., Hossen, M.R., & Ferdous, J. Android Based Low Cost Sitting Posture Monitoring System. *IEEE ICECE*, 2020. [Paper]

Invited Talks & Presentations

Shapley-Adaptive Modality Knockout for Robust Multimodal Learning. 2026
MICCAI, Strasbourg, France

X-Cardia: Phenotype-Guided Cross-Modal Alignment for Cardiac Screening 2026
Medical Imaging with Deep Learning (MIDL), Taipei, Taiwan

Multimodal Deep Learning for Opportunistic Cardiac Amyloidosis Screening 2025
Young Investigators' Award Competition, ACC New York Chapter, New York, NY

Opportunistic Screening of Cardiac Amyloidosis 2025
Cardiovascular Research Symposium, Weill Cornell Medicine, New York, NY

Honors & Awards

1st Runner-up, ACT COVID-19 National Call Hackathon, ICT Ministry	2020
1st Runner-up, Johns Hopkins Healthcare Design Competition	2019
1st Place, EEE Day Inter-University Project Show, BUET	2019
2nd Best Presenter Award, Bangladesh Cardiac Society Conference	2019
2nd Runner-up, Design for Life Competition, Edge Foundation, Bangladesh	2019
Dean's List Scholarship — all academic levels of undergraduate study, BUET	2016 – 2020
University Merit Scholarship — multiple academic terms, BUET	2016 – 2020

Leadership & Service

Student Mentor, Cornell Tech 2026 – Present
Mentor and guide undergrad students in different research projects.

International Student Advisory Committee, Cornell Tech 2024 – 2026
Represent international graduate students in policy and student-welfare discussions

Co-founder, GyanJam — educational technology platform 2018 – 2020
Built and ran an online learning platform reaching many learners in Bangladesh